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Essentials of Data Science Laboratory - 2304102L - 2026

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1.1.1. Calculate Momentum

150%

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum  $p$  is calculated using the formula:  
$$p = m \times v$$
  
where:  
 $m$  is the mass of the object (in kilograms).  
 $v$  is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

Submit

1#Write a python program to calculate momentum

2

3mass = float(input())

4velocity = float(input())

5

6momentum = mass \* velocity

7print(f"momentum: .2f)kgm/s")

Average time0.024 s23.50 ms

Maximum time0.068 s68.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 168 msDebug

Expected output

5.0

10.0

50.00kgm/s

Actual output

5.0

10.0

50.00kgm/s

Test case 29 ms

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1.1.2. Conditional Calculation Based on the Number o...12.07

Write a Python program that accepts an integer  $n$  as input. Depending on the number of digits in  $n$ .

Constraints:  
 $1 \leq n \leq 999$

Input Format:  
• The input consists of a single integer  $n$ .

Output Format:  
• If  $n$  is a single-digit number, print its square.  
• If  $n$  is a two-digit number, print its square root (rounded to two decimal places).  
• If  $n$  is a three-digit number, print its cube root (rounded to two decimal places).  
• Else print "Invalid".

Sample Test Cases

condition...

Submit

1#write your code here...  
2  
3n = int(input())  
4  
5count = len(str(n))  
6  
7if count == 1:  
8    print(n \* n)  
9elif count == 2:

Average time0.005 s5.43 ms  
Maximum time0.008 s8.00 ms

4 out of 4 shown test case(s) passed  
3 out of 3 hidden test case(s) passed

Test case 18 msDebug

Expected output9  
81  
Actual output9  
81

Test case 24 ms

Test case 35 ms

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1.1.3. Days between Two Dates00:00

Write a Python program to calculate number of days between two dates.

Input Format:

The first line contains the first date in the format YYYY – MM – DD.

The second line contains the second date in the format YYYY – MM – DD.

Output Format:

An integer representing the number of days between the given dates.

Note:

The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

1from datetime import date

2

3#write your code here...

4d1 = input().split("-")

5d2 = input().split("-")

6

7date1 = date(int(d1[0]), int(d1[1]), int(d1[2]))

8date2 = date(int(d2[0]), int(d2[1]), int(d2[2]))

9

Average time0.038 s37.75 ms

Maximum time0.078 s78.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 139 ms

Expected output2014-07-022014-11-02123

Actual output2014-07-022014-11-02123

Test case 216 ms

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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 #write your code here...
2
3 n = int(input())
4 rev = int(str(n)[::-1])
5 print(rev)
```

Average time0.011 s11.20 ms

Maximum time0.014 s14.00 ms

2 out of 2 shown test case(s) passed3 out of 3 hidden test case(s) passed

Test case 112 ms

Expected output53677635

Actual output53677635

Test case 211 ms

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1.1.5. Multiplication Table04:12

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

The character "x" is used in the output to represent multiplication.  
Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

Submit

1#write your code here...

2

3n = int(input())

4

5for i in range(1,11):

6    print(n, 'x', i, "=", n \* i)

7

Average time:0.009 s8.50 ms

Maximum time:0.012 s12.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 112 msDebug

Expected output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

Actual output

8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

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44/46

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

- The first input will be an integer  $n$ , the number of courses.
- The second input will be  $n$  integers representing the marks of the student in each of the  $n$  courses, separated by a space.

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Submit

```
1 # Input: number of courses
2 n = int(input())
3
4 # Input: marks
5 marks = list(map(int, input().split()))
6
7 # Check for fail condition
8 failed = False
9 for m in marks:
```

Average time: **0.014 s** (14.30 ms) Maximum time: **0.018 s** (18.00 ms)

5 out of 5 shown test case(s) passed  
5 out of 5 hidden test case(s) passed

|                             |                             |
|-----------------------------|-----------------------------|
| 4                           | 4                           |
| 55 65 78 89                 | 55 65 78 89                 |
| Aggregate Percentage: 71.75 | Aggregate Percentage: 71.75 |
| Grade: First Division       | Grade: First Division       |

Test case 2 10 ms

Test case 3 10 ms

Terminal Test cases



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1.2.2. Fibonacci Series

00:35

Write a Python program that uses recursion to print the first  $n$  terms of the Fibonacci series.

Input Format:

A single integer  $n$  representing the number of terms to generate.

Output Format:

A single line containing the first  $n$  terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

Submit

1def fibonacci(n):

2# base cases

3if n == 1:

4return 0

5elif n == 2:

6return 1

7else:

8return fibonacci(n-1) + fibonacci(n-2)

9

Average time:0.035 s34.75 ms

Maximum time:0.096 s96.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 16 ms

Debug

Expected output

5

0 1 1 2 3

Actual output

5

0 1 1 2 3

Test case 212 ms

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1.2.3. Pattern - 1

1.2.4. Pattern - 2

2. Practical 2

3. Practical 3

4. Practical 4

5. Practical 5

1.2.3. Pattern - 101:15

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

The output should display the pattern of asterisks (\*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

Submit

1n = int(input())

2

3for i in range(1, n + 1):

4for j in range(i):

5print("\*", end=" ")

6print()

Average time0.014 s14.33 ms

Maximum time0.017 s17.00 ms

2 out of 2 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 117 msDebug

Expected output

5

\*

\* \*

\* \* \*

\* \* \* \*

\* \* \* \* \*

Actual output

5

\*

\* \*

\* \* \*

\* \* \* \*

\* \* \* \* \*

Terminal

Test cases

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1.2.3. Pattern - 1

1.2.4. Pattern - 2

2. Practical 2

3. Practical 3

4. Practical 4

5. Practical 5

1.2.4. Pattern - 202/24

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

Submit

1n = int(input())

2

3for i in range(1, n + 1):

4    for j in range(1, i + 1):

5        print(j, end=" ")

6    print()

7

8

Average time0.019 s19.25 ms

Maximum time0.029 s29.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 129 msDebug

Expected output

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Actual output

5

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Terminal

Test cases

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